

CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 8 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. To decide the winner in a League tournament, the 'British Method' uses which of the following mathematical formulas? (A) $\frac{\text{Total Matches Won}}{\text{Total Matches Played}} \times 100$ (B) $\frac{\text{Total Points Obtained}}{\text{Total Possible Points}} \times 100$ (C) $\frac{\text{Total Matches Played}}{\text{Total Points}} \times 100$ (D) $\frac{\text{Total Points Obtained}}{\text{Total Matches Played}} \times 100$ Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High**

Q2. The postural deformity characterized by an abnormal inward dropping of the knees, causing them to touch or overlap while standing, is medically known as: (A) Genu Varum (B) Genu Valgum (C) Pes Planus (D) Kyphosis Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High**

Q3. Which of the following Pranayamas is famously known as the "Cooling Breath" and helps in reducing body temperature and high blood pressure? (A) Anulom Vilom (B) Kapalbhati (C) Sitli Pranayama (D) Suryabhedan Pranayama Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [Practice] | Tag: High**

Q4. The International Committee of Sports for the Deaf (ICSD) is the officially recognized governing body for which mega sporting event? (A) Special Olympics (B) Paralympics (C) Deaflympics (D) Commonwealth Games Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: Very High**

Q5. Which of the following is strictly classified as a 'Non-Nutritive' component of the human diet? (A) Proteins (B) Minerals (C) Colouring Compounds (D) Vitamins Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High**

Q6. In the SAI Khelo India Fitness Test battery for boys aged 9-18 years, the 'Push-up' test is conducted to measure: (A) Cardiovascular endurance (B) Core strength (C) Upper body strength and endurance (D) Explosive power Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High**

Q7. The maximum volume of air that can be voluntarily exhaled after a maximal, deep inhalation is called: (A) Tidal Volume (B) Residual Volume (C) Minute Volume (D) Vital Capacity Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High**

Q8. A push-up or a calf raise (standing on tiptoes) is a biomechanical example of which class of lever in the human body? (A) Class I Lever (B) Class II Lever (C) Class III Lever (D) Class IV Lever Unit 8: Biomechanics and Sports | [PYQ] | Tag: Very High**

Q9. Which organizing committee is responsible for arranging medals, trophies, and certificates for the winners during the closing stages of a sports event? (A) Reception

Committee (B) Decoration and Ceremony Committee (C) Transport Committee (D) Technical Committee *Unit 1: Management of Sporting Events | [Practice] | Tag: High***

Q10 **শারীরিক**. According to the World Health Organization (WHO) guidelines, children and youth aged 5-17 years should accumulate at least how many minutes of moderate-to-vigorous physical activity daily? (A) 30 minutes (B) 45 minutes (C) 60 minutes (D) 120 minutes *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***

Q11. Which of the following lifestyle diseases is fundamentally a metabolic disorder where the pancreas fails to produce sufficient insulin? (A) Asthma (B) Hypertension (C) Obesity (D) Diabetes *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: Very High***

Q12. The Paralympic Games are held immediately following the Olympic Games in the same host city. Who are the primary participants in the Paralympics? (A) Athletes with cognitive and intellectual disabilities (B) Athletes with physical, visual, and intellectual impairments (C) Athletes with hearing impairments (D) Veteran athletes *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: Very High***

Q13. Phosphorus, Magnesium, and Potassium are required by the body in larger amounts and are therefore classified as: (A) Micro Minerals (B) Trace Minerals (C) Macro Minerals (D) Non-nutritive minerals *Unit 5: Sports & Nutrition | [PYQ] | Tag: High***

Q14. In the mathematical formula to calculate the Body Mass Index (BMI), a person's body weight is measured in kilograms and their height is measured in: (A) Centimeters (B) Inches (C) Meters squared (m^2) (D) Feet squared *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q15. A severe sports injury characterized by the overstretching or tearing of a ligament at a joint is known as a: (A) Strain (B) Sprain (C) Contusion (D) Abrasion *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

Q16. Assertion (A): In throwing events like javelin or shot put, an athlete's 'follow-through' motion is highly critical after the release. **Reason (R):** A proper follow-through abruptly and immediately stops the athlete's body motion to prevent them from crossing the foul line. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports | [Practice] | Tag: High***

Q17. Assertion (A): Vitamins and minerals are categorized as nutritive components of a diet. **Reason (R):** Vitamins and minerals provide high amounts of caloric energy necessary for intense sports performance. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***

Q18. Assertion (A): Extramural sports competitions significantly raise the standard of sports in participating schools. **Reason (R):** Extramural tournaments provide students the opportunity to compete with highly skilled players from other institutions, exposing them to superior techniques and tactics. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true,

but (R) is false. (D) (A) is false, but (R) is true. *Unit 1: Management of Sporting Events* | [PYQ] | Tag: Very High**

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

Q19. Mention any two distinct disadvantages of a League Tournament. *Unit 1: Management of Sporting Events* | [PYQ] | Tag: High**

Q20. State any two physical corrective measures or exercises for the postural deformity known as 'Kyphosis' (Hunchback). *Unit 2: Children & Women in Sports* | [PYQ] | Tag: Very High**

Q21. Enlist any two physiological or mental benefits of practicing *Shavasana* (Corpse Pose). *Unit 3: Yoga as Preventive Measure for Lifestyle Disease* | [PYQ] | Tag: High**

Q22. What are the primary objectives of the 'Special Olympics' movement? *Unit 4: Physical Education & Sports for CWSN* | [PYQ] | Tag: High**

Q23. Briefly explain the purpose and the specific equipment required for conducting the 'Sit and Reach Test'. *Unit 6: Test and Measurement in Sports* | [PYQ] | Tag: Very High**

Q24. Define 'Cardiac Output'. Write its simple mathematical formula. *Unit 7: Physiology & Injuries in Sport* | [Practice] | Tag: Very High**

SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

Q25. Define Intramural tournaments. Write down any two points highlighting the significance of organizing intramurals in schools. *Unit 1: Management of Sporting Events* | [PYQ] | Tag: Very High**

Q26. Elaborate on the Physical Exercise Guidelines recommended by the World Health Organization (WHO) for children under 5 years of age. *Unit 2: Children & Women in Sports* | [PYQ] | Tag: High**

Q27. "Water is a non-nutritive component of diet, yet it is essential for an athlete." Justify this statement with three distinct physiological reasons. *Unit 5: Sports & Nutrition* | [PYQ] | Tag: Very High**

Q28. Describe the purpose and the step-by-step administration procedure of the '8-Foot Up and Go Test' for Senior Citizens. *Unit 6: Test and Measurement in Sports* | [PYQ] | Tag: Very High**

Q29. Regular exercise delays the onset of physical decline. Elucidate any three major physiological changes that naturally occur in the human body due to the process of aging. *Unit 7: Physiology & Injuries in Sport* | [PYQ] | Tag: High**

Q30. Define 'Friction'. Differentiate clearly between Static Friction and Dynamic (Sliding/Rolling) Friction, giving one sports example of each. *Unit 8: Biomechanics and Sports* | [PYQ 2024] | Tag: Very High**

SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

Q31. XYZ Public School is organizing an Inter-School Hockey Championship. The Physical Education teacher decides to conduct the tournament on a Single League basis using the 'Cyclic Method'. There are 6 teams participating in total. The sports captain is tasked with calculating the fixture details before drawing it. **(i)** Write the mathematical formula used to calculate the total number of matches in a Single League tournament. **(ii)** Calculate the total number of rounds that will be played in this 6-team cyclic fixture. **(iii)** Will any 'Bye' be given in this specific fixture? (Yes/No) **(iv)** What is the primary advantage of a league tournament format? **OR** Write the formula to calculate total matches if the tournament was played on a Double League basis. *Unit 1: Management of Sporting Events* | [Practice] | Tag: Very High**

Q32. During an intense inter-house basketball match, a player trips and slides across the rough outdoor concrete court, badly scraping his knee. Later in the same game, he violently collides with a heavy defender and gets a painful blue/black swelling on his outer thigh. The school medical team immediately attends to him. **(i)** The surface-level scraping of the skin on his knee is medically termed as a/an __. **(ii)** The blue/black swelling on his thigh, caused by internal bleeding without breaking the skin, is known as a/an __. **(iii)** Both of these injuries are classified under which broad category of sports injuries? (Bone / Joint / Soft Tissue) **(iv)** What immediate first-aid protocol should the medical team apply to treat the swelling on his thigh? **OR** Define a 'Laceration'. *Unit 7: Physiology & Injuries in Sport* | [PYQ] | Tag: Very High**

Q33. A Javelin thrower is practicing in the field to qualify for the national meet. His biomechanics coach is recording his throws on an iPad. The coach notes the athlete's speed of run-up, the height at which the javelin leaves his hand, and the angle of release. The coach explains that manipulating these factors will maximize the horizontal distance the javelin travels in the air. **(i)** The parabolic path followed by the javelin in the air is scientifically termed as its __. **(ii)** What is the theoretically optimum angle of release to achieve the maximum possible horizontal distance? **(iii)** Out of angle, height, and speed of release, which factor is considered the most critical in determining the distance of the projectile? **(iv)** Name the invisible downward force that pulls the javelin back toward the ground. **OR** Define a 'Projectile' in sports. *Unit 8: Biomechanics and Sports* | [PYQ] | Tag: High**

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 5)

Word limit: 200-300 words.

Q34. The success of a sports tournament heavily relies on effective committee management. Thoroughly explain the distinct Pre-tournament, During-tournament, and Post-tournament responsibilities of the Organizing Committees. *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q35. The 'Female Athlete Triad' is a severe syndrome affecting women in sports. Define the Triad. Comprehensively explain its three interconnected components (Eating Disorders, Amenorrhea, Osteoporosis) and how they negatively impact a female athlete's health. *Unit 2: Children & Women in Sports | [PYQ 2024] | Tag: Very High***

Q36. Define 'Asthma'. Name any four Yoga Asanas prescribed in the syllabus for the prevention and management of Asthma. Describe the step-by-step procedure, two benefits, and two contraindications of *Matsyasana* (Fish Pose). *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: Very High***

Q37. What are 'Bone Injuries' in sports? Classify and explain any four types of bone fractures (e.g., Simple, Compound, Comminuted, Greenstick, Transverse, Oblique). Outline the general immediate management (first-aid) for a suspected bone fracture on the field. *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: High***

Answer Key

Q1. (B) $\frac{\text{Total Points Obtained}}{\text{Total Possible Points}} \times 100$ **Q2.** (B) Genu Valgum **Q3.** (C) Sitli Pranayama **Q4.** (C) Deaflympics **Q5.** (C) Colouring Compounds **Q6.** (C) Upper body strength and endurance **Q7.** (D) Vital Capacity **Q8.** (B) Class II Lever **Q9.** (B) Decoration and Ceremony Committee **Q10.** (C) 60 minutes **Q11.** (D) Diabetes **Q12.** (B) Athletes with physical, visual, and intellectual impairments **Q13.** (C) Macro Minerals **Q14.** (C) Meters squared (m^2) **Q15.** (B) Sprain **Q16.** (C) (A) is true, but (R) is false. (Reason is false because a proper follow-through ensures smooth and gradual deceleration, preventing jerky motions and injuries, rather than stopping abruptly). **Q17.** (C) (A) is true, but (R) is false. (Reason is false because vitamins and minerals provide 0 calories/energy). **Q18.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

Q19. 1. **Time-Consuming:** It requires a large number of matches, making it a very long tournament. 2. **Expensive:** More matches mean higher costs for organizing, officiating, and boarding. **Q20.** 1. Performing *Chakrasana* (Wheel Pose) or *Dhanurasana* (Bow Pose) to stretch the chest muscles. 2. Swimming or sitting on a chair with a pillow behind the mid-back and bending backward. **Q21.** 1. Relaxes the entire nervous system and cures mental stress, anxiety, and depression. 2. Regulates blood pressure and cures insomnia by resting the body completely. **Q22.** The primary objective of the Special Olympics is to provide year-round sports training and athletic competition in various Olympic-type sports for children and adults strictly with **intellectual/cognitive disabilities**, helping them develop physical fitness and integrate into society. **Q23. Purpose:** To measure the flexibility of the lower back and hamstring muscles. **Equipment:** A specially constructed wooden 'Sit and Reach' box with a

measurement scale (in cm) extending over the top. **Q24. Cardiac Output** is the total volume of blood pumped out by the heart (left ventricle) in one minute. **Formula:**

$$\text{CardiacOutput} = \text{HeartRate} \times \text{StrokeVolume}.$$

Q25. Intramural Tournaments are sports competitions organized strictly within the boundaries or walls of an institution, involving only the students of that school. **Significance:**

1. It provides opportunities for mass participation, ensuring every child plays. 2. It helps discover hidden sports talent within the school. **Q26. WHO Guidelines (< 5 years):** 1. Infants should be physically active several times a day (e.g., floor-based play/tummy time). 2.

Toddlers (1-4 years) should spend at least 180 minutes in various physical activities spread throughout the day. 3. Screen time (TV/phones) is strongly not recommended for 1-year-olds and limited to <1 hour for 2-4-year-olds. **Q27. 1. Thermoregulation:** Water regulates body

core temperature through sweating during intense sports. 2. **Nutrient Transport:** It acts as a fluid medium in the blood to transport oxygen, glucose, and electrolytes to the working muscles. 3. **Waste Removal:** It flushes out metabolic waste products (like uric acid and urea) through urine. **Q28. Purpose:** To measure agility and dynamic balance in older adults.

Procedure: The participant sits in a chair with hands resting on knees. On the "Go" command, they stand up, walk as quickly as possible around a cone placed 8 feet away, return to the chair, and sit down. The time taken is recorded in seconds. **Q29. 1. Loss of Muscle Mass (Sarcopenia):** Muscular size and strength visibly decrease. 2. **Reduced Bone Density:** Bones lose calcium, becoming brittle and prone to fractures. 3. **Decreased Lung Capacity:** The elasticity of lung tissues decreases, reducing vital capacity and oxygen uptake.

Q30. Friction is the opposing force that resists the movement of one surface over another.

Static Friction: The friction present when the object is completely at rest (e.g., a sprinter's spikes gripping the starting blocks before the gun fires). **Dynamic Friction:** The friction present when the object is actively moving/sliding/rolling (e.g., a hockey ball rolling across turf).

Q31. (i) Total Matches = $\frac{N(N-1)}{2}$. (ii) Rounds = $N - 1 = 6 - 1 = 5$ rounds. (iii) No (Since $N = 6$ is an even number, no byes are given in cyclic method). (iv) It ensures that every team gets a chance to play against all other teams regardless of early losses, determining the true best team. **OR** Formula for Double League: $N(N - 1)$. **Q32.** (i) Abrasion. (ii) Contusion. (iii) Soft Tissue Injuries. (iv) Use the **PRICE** protocol (Protect, Rest, Ice, Compress, Elevate). Apply an ice pack immediately to constrict blood vessels and reduce swelling. **OR** A Laceration is an irregular, jagged tear in the skin tissue caused by blunt trauma or a rough impact. **Q33.** (i) Trajectory (or Parabola). (ii) 45° . (iii) Initial Velocity (Speed of release). (iv) Gravity. **OR** A Projectile is any object thrown or projected into the air that continues its motion entirely under the influence of gravity (and air resistance).

Q34. Organizing Committees divide workloads. **Pre-Tournament Responsibilities:** 1. Preparing the event budget and securing sponsorships. 2. Sending invitations to schools and receiving entries. 3. Preparing the grounds, tracks, and purchasing necessary sports equipment. 4. Drawing up the official tournament fixtures and schedules. **During-Tournament Responsibilities:** 1. Checking the eligibility of participating teams and

conducting matches strictly as per the schedule. 2. Providing first-aid immediately in case of injuries. 3. Managing the officiating panel (referees, umpires) and ensuring fair play. 4. Providing food and accommodation to athletes.

Post-Tournament Responsibilities: 1. Organizing the closing ceremony and distributing medals/trophies to winners. 2. Preparing and presenting detailed financial expenditure reports to the administration. 3. Returning borrowed equipment and cleaning the sports venue.

Q35. The Female Athlete Triad is a serious health syndrome seen in female athletes driven by extreme physical training and poor dietary habits. It consists of three interrelated disorders: 1. **Eating Disorders:** Often the starting point. Due to the psychological pressure to maintain a very low body weight for aesthetic sports (gymnastics) or weight-class sports (wrestling), athletes intentionally starve themselves (Anorexia) or binge and purge (Bulimia). This leads to severe "low energy availability." 2. **Amenorrhea:** The absence of menstrual periods for three or more consecutive months. The intense physical stress combined with drastic caloric deficiency signals the brain to shut down non-essential functions like reproduction, causing estrogen levels to plummet. 3. **Osteoporosis:** Severe weakening of the bones. Estrogen is critical for bone calcium absorption. Without estrogen (due to amenorrhea) and lacking dietary calcium, bones become highly porous, brittle, and extremely susceptible to stress fractures during sports. **Q36. Asthma** is a chronic inflammatory disease of the respiratory airways. The airways swell and produce extra mucus, causing wheezing and severe shortness of breath.

Recommended Asanas: Matsyasana, Chakrasana, Gomukhasana, Bhujangasana.

Matsyasana (Fish Pose) Procedure: 1. Sit in Padmasana (Lotus pose). 2. Lean backward and slowly lie flat on the back with the support of the elbows. 3. Lift the chest upward, arch the back, and rest the crown of the head on the floor. 4. Hold the big toes with your hands. Keep breathing deeply. **Benefits:** 1. Greatly expands the chest cage and lungs, increasing oxygen capacity and directly curing asthma. 2. Cures postural deformities like round shoulders.

Contraindications: 1. Individuals suffering from severe neck/cervical pain or back injuries should avoid it. 2. People with high blood pressure or hernia must not perform this pose.

Q37. A Bone Fracture is a partial or complete break in the continuity of a bone due to high impact. **Types:** 1. *Simple (Closed) Fracture:* The bone breaks but the skin remains intact. 2. *Compound (Open) Fracture:* The broken bone splinters and violently pierces the skin, exposing the bone and posing a high infection risk. 3. *Comminuted Fracture:* The bone is crushed into three or more pieces due to a heavy smash. 4. *Greenstick Fracture:* The bone bends and cracks without breaking into separate pieces (common in children's softer bones). 5. *Transverse Fracture:* A straight break directly across the long axis of the bone. **Immediate Management:** 1. Do not move the victim or try to align/pop the broken bone back into place. 2. Completely **immobilize** the injured area immediately using a rigid splint, sling, or bandages. 3. Control any external bleeding with a clean cloth (in case of a compound fracture). 4. Apply ice packs wrapped in cloth to reduce swelling. 5. Call for emergency medical help or transport to a hospital.